

Foundation Program in Literacy, Numeracy and Skills

NUMERACY Teacher's guide Grade 8: Lesson 3

A. Multiplication and Division

B. Competency, Element and Performance Criteria:

C 2.5: Solving simple mathematical problems in practical situations

Element 1: Solve problems involving numbers in real life contexts

Performance Criteria: -

Level 1: Apply simple mathematical operations in familiar real-life situations.

Level 2: Solve multi-step problems involving numbers in real life contexts.

C. Purpose of activity:

To enhance students' skills in two digit multiplication and division through real life shopping scenarios.

D. Learning Outcomes:

By the end of the lesson, students should be able to

- Multiply and divide by 2-digit numbers.
- Apply multiplication and division in real-life shopping.
- Interpret word problems and choose appropriate operations.

E. Resources and materials:

- Supermarket brochures or items/pictures with price tags.

F. Teaching trajectories/Implementation guidelines:

Teacher to ensure that students are able to carry out multiplication and division by 2-digit numbers using algorithms before students can explore the use of multiplication and division on the theme shopping.

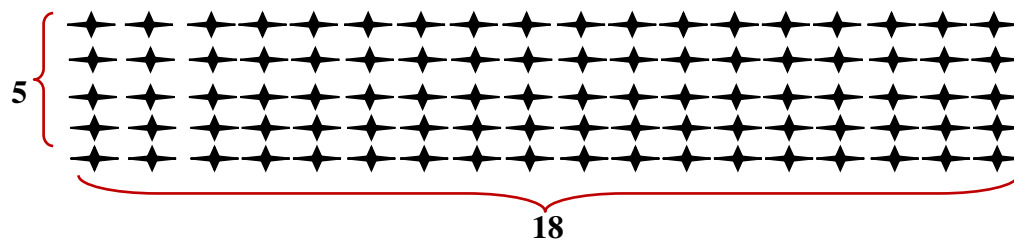
Multiplication of a 2-digit number by a 1-digit number

Example: Carry out the multiplication 18×5 by different ways (repeated addition, array method and by grouping).

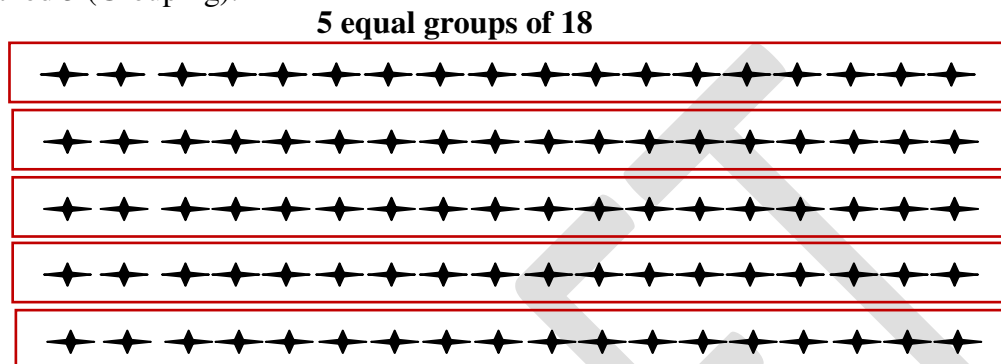
Method1 (Repeated addition): $18 \times 5 = 18 + 18 + 18 + 18 + 18$

$$= 90$$

Method 2 (Array Method):



Method 3 (Grouping):



Multiplication by 2-digit numbers

Example: Let us consider 63×94

Algorithm:

1. Multiply 4 (units) by 63. (Pretend the 9 in 94 is not there.)
2. Then multiply 90 (9 tens) by 63, but put the result underneath the 252. Remember the zero. (Pretend the 4 in 94 is not there.)
3. Then add both lines. ($252 + 5670$)

4×63	90×63	Add.
$\begin{array}{r} 1 \\ 63 \\ \times 94 \\ \hline 252 \end{array}$	$\begin{array}{r} 2 \\ 63 \\ \times 94 \\ \hline 252 \\ 5670 \end{array}$	$\begin{array}{r} 63 \\ \times 94 \\ \hline 252 \\ + 5670 \\ \hline 5922 \end{array}$

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NUMERACY Teacher's guide Grade 8: Lesson 4

Division by 2-digit numbers using long division

Example: Let us consider $5568 \div 16$

Often, it is helpful to write the multiplication table of the divisor before you divide.			
Example 1. The division is by 16. Here is the multiplication table of 16: $3 \times 16 = 48$ $4 \times 16 = 64$ $5 \times 16 = 80$ $6 \times 16 = 96$ $7 \times 16 = 112$ $8 \times 16 = 128$ $9 \times 16 = 144$	$\begin{array}{r} 03 \\ 16 \overline{) 5568} \\ \underline{48} \\ 76 \end{array}$ 16 goes into 5 zero times, so we look at 55. How many times does 16 go into 55? Check in the table on the left. We see it goes into 55 three times.	$\begin{array}{r} 034 \\ 16 \overline{) 5568} \\ \underline{48} \\ 76 \end{array}$ Now, how many times does 16 go into 76? From the table we can see that it is four times.	$\begin{array}{r} 0348 \\ 16 \overline{) 5568} \\ \underline{48} \\ 76 \\ \underline{64} \\ 128 \\ \underline{128} \\ 0 \end{array}$ Lastly, 16 goes into 128 exactly 8 times, and the division is over.

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NUMERACY Teacher's guide Grade 8: lesson 5

Activity 1: Group task on bulk shopping

Scenario: You are shopping for a school event. Items must be bought in bulk.

- Example 1 (Multiplication):

You want to buy 15 packs of paper plate. Each pack cost Rs 56. What is the total cost?

- Example 2 (Division):

You spent Rs 780 On 30 gift boxes. What is the cost of one gift box?

Activity 2: Shopping Game (Group Activity)

- Provide a shopping catalog with item prices.
- Give tasks like:
 - “How much will it cost to buy 28 T-Shirts at Rs 75 each?”
 - “You have Rs 1240. Can you buy 20 lunch boxes at Rs 62 each?”

H. Extension of activity

- Discount and percentage discount.

[Discuss with students: “Shopkeepers sometimes offer goods at reduced prices in order to dispose old or damaged goods or to increase sales or to attract customers to buy their product”]